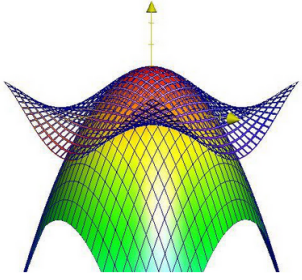
Optimization in Machine Learning

First order methods
Weaknesses of GD – Curvature



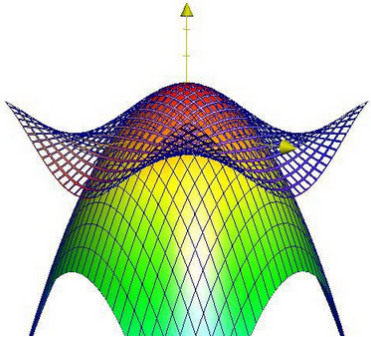
Learning goals

- Effects of curvature
- Step size effect in GD

REMINDER: LOCAL QUADRATIC GEOMETRY

Locally approximate smooth function by quadratic Taylor polynomial:

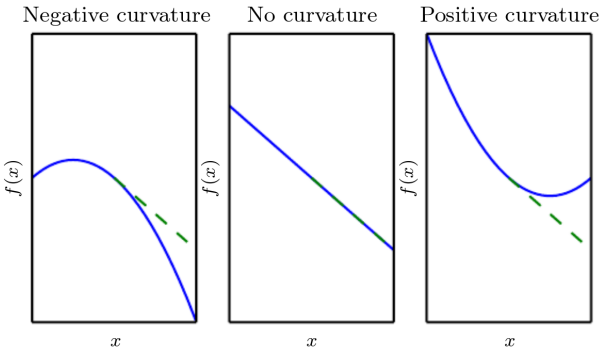
$$f(\mathbf{x}) \approx f(\tilde{\boldsymbol{x}}) + \nabla f(\tilde{\boldsymbol{x}})^\top (\mathbf{x} - \tilde{\boldsymbol{x}}) + \frac{1}{2}(\mathbf{x} - \tilde{\boldsymbol{x}})^\top \nabla^2 f(\tilde{\boldsymbol{x}})(\mathbf{x} - \tilde{\boldsymbol{x}})$$



Source: `daniloroccatano.blog`.

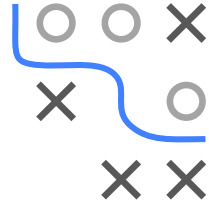
EFFECTS OF CURVATURE

Intuitively, curvature determines reliability of a GD step



Quadratic objective f (blue) with gradient approximation (dashed green). **Left:** f decreases faster than ∇f predicts. **Center:** ∇f predicts decrease correctly. **Right:** f decreases more slowly than ∇f predicts. (Source: Goodfellow et al., 2016)

CURVATURE AND STEP SIZE IN GD



Worst case: H is ill-conditioned. What does this mean for GD?

- Quadratic Taylor polynomial of f around $\tilde{\mathbf{x}}$ (with gradient $\mathbf{g} = \nabla f$)

$$f(\mathbf{x}) \approx f(\tilde{\mathbf{x}}) + (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{g} + \frac{1}{2}(\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{H}(\mathbf{x} - \tilde{\mathbf{x}})$$

- GD step with step size $\alpha > 0$ yields

$$f(\tilde{\boldsymbol{x}} - \alpha \mathbf{g}) \approx f(\tilde{\boldsymbol{x}}) - \alpha \mathbf{g}^\top \mathbf{g} + \frac{1}{2}\alpha^2 \mathbf{g}^\top \boldsymbol{H} \mathbf{g}$$

- If $\mathbf{g}^\top \boldsymbol{H}\mathbf{g} > 0$, we can solve for optimal step size α^*:

$$\alpha^* = \frac{\mathbf{g}^\top \mathbf{g}}{\mathbf{g}^\top \boldsymbol{H} \mathbf{g}}$$